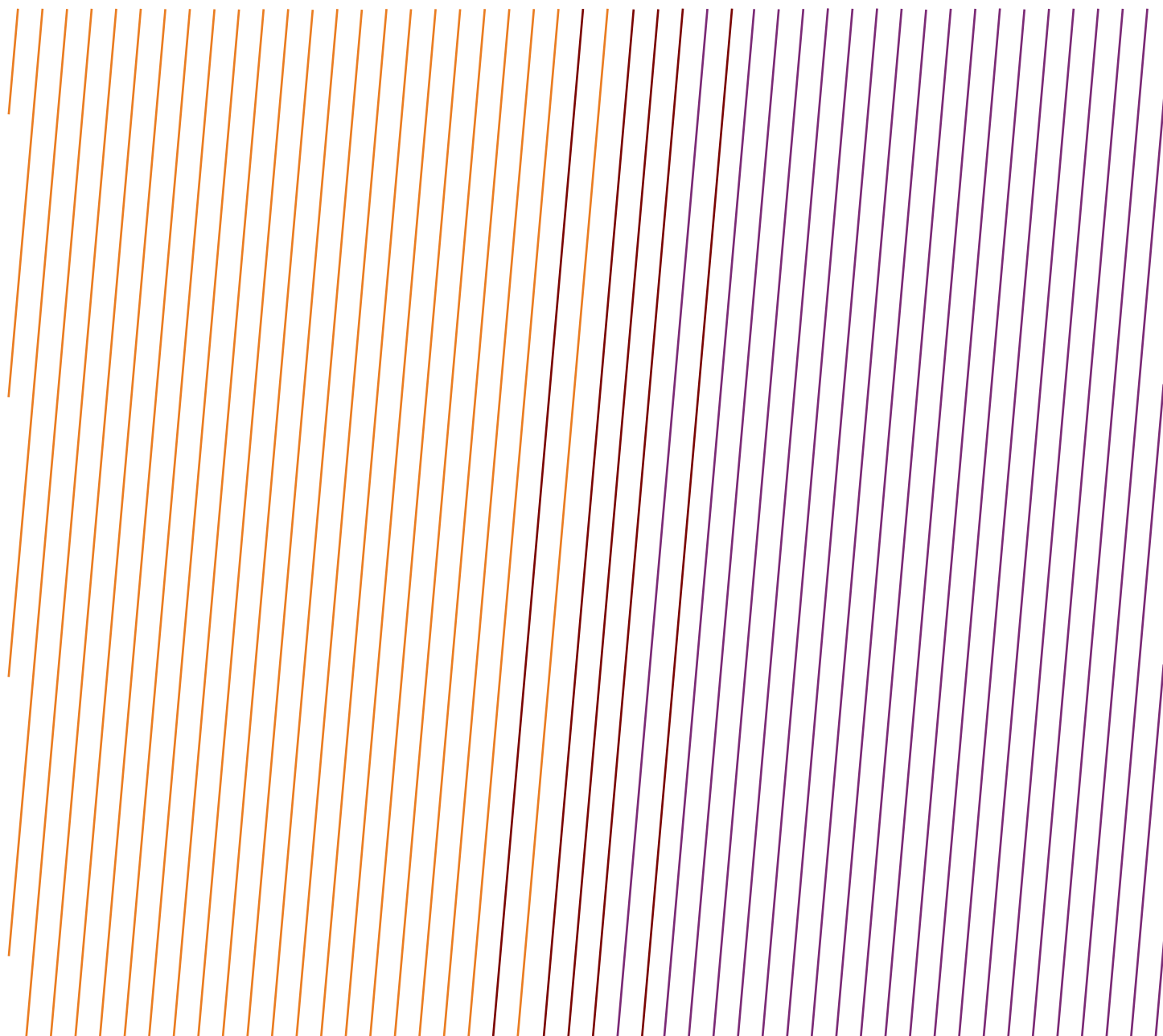


DATA SERIES

Safety performance indicators – Permanent impairment incident reports – 2024



Acknowledgements

IOGP thanks those companies that have participated in the data collection programme.

This Report was produced by the Safety Committee.

Feedback

IOGP welcomes feedback on our reports: publications@iogp.org

Disclaimer

Whilst every effort has been made to ensure the accuracy of the information contained in this publication and any associated material, neither IOGP nor any of its Members past present or future warrants its accuracy or will, regardless of its or their negligence, assume liability for any foreseeable or unforeseeable use made thereof, which liability is hereby excluded. Consequently, such use is at the recipient's own risk on the basis that any use by the recipient constitutes agreement to the terms of this disclaimer. The recipient is obliged to inform any subsequent recipient of such terms.

Please note that this publication and any associated material is provided for informational purposes and adoption of any of its recommendations is at the discretion of the user. Except as explicitly stated otherwise, this publication must not be considered as a substitute for government policies or decisions or reference to the relevant legislation relating to information contained in it.

Where the publication contains a statement that it is to be used as an industry standard, IOGP and its Members past, present, and future expressly disclaim all liability in respect of all claims, losses or damages arising from the use or application of the information contained in this publication in any industrial application.

Any reference to third party names is for appropriate acknowledgement of their ownership and does not constitute a sponsorship or endorsement.

Copyright notice

This publication and the contents of these pages are © International Association of Oil & Gas Producers™ ("IOGP™"). Permission is given to reproduce, quote from, republish, redistribute, and recommunicate this publication in whole or in part for purposes without independent economic value [2] and provided that (i) International Association of Oil & Gas Producers is always acknowledged as the source, together with the words "© International Association of Oil & Gas Producers™ ("IOGP™"), and (ii) any acknowledgments of other sources as may appear in this publication, in respect of such material, are included unaltered. All other rights are reserved.

These Terms and Conditions shall be governed by and construed in accordance with the laws of England and Wales. Disputes arising here from shall be exclusively subject to the jurisdiction of the courts of England and Wales.

Safety performance indicators – Permanent impairment incident reports – 2024

Revision history		
VERSION	DATE	AMENDMENTS
1.0	December 2025	First release

Contents

IOGP permanent impairment incidents 2024	5
AFRICA ONSHORE	5
AFRICA OFFSHORE	9
ASIA/AUSTRALASIA ONSHORE	10
ASIA/AUSTRALASIA OFFSHORE	12
EUROPE ONSHORE	13
EUROPE OFFSHORE	14
MIDDLE EAST ONSHORE	17
MIDDLE EAST OFFSHORE	25
NORTH AMERICA ONSHORE	26
NORTH AMERICA OFFSHORE	34
RUSSIA & CENTRAL ASIA ONSHORE	35
RUSSIA & CENTRAL ASIA OFFSHORE	36
SOUTH & CENTRAL AMERICA ONSHORE	36
SOUTH & CENTRAL AMERICA OFFSHORE	38

IOGP permanent impairment incidents 2024

AFRICA ONSHORE

DATE: Aug 8 2024

COUNTRY: Egypt

WORKFORCE PI INJURIES: 1

FUNCTION: Drilling

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Drilling, workover, well operations

PRIMARY LIFE-SAVING RULE: Line of fire

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent loss of body parts

PI Subcategory: Amputations with loss of bone inclusive of one or more joints (not able to be reattached)

LWDC Days: -

RWDC Days: -

Function: Drilling

Employer: Contractor

Occupation: Drilling/well servicing operator

Body part: Finger(s)/thumb

Nature of injury: Amputation

Time in service: Unspecified

NARRATIVE:

The operation was to make up the ESP assembly. One of the rig crew (IP) was rolling one section of ESP pump on the pipe rack to the catwalk in order to be lifted to rig floor, and suddenly his right hand was jammed between the catwalk and the pump, causing two cut wounds on his right hand fingers phalanges. Injury resulted in amputation of both fingers at the distal phalange.

WHAT WENT WRONG:

Improper position for task and failure to utilize proper pipe handling tools/equipment.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Training on proper use of pipe handling tools will be pushed to applicable work crews. Use of pipe handling equipment required when moving piping on racks.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: Nov 26 2024
COUNTRY: Libya
WORKFORCE PI INJURIES: 1
FUNCTION: Drilling
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Energy isolation
SECONARY LIFE-SAVING RULE: Line of fire

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent loss of body parts
PI Subcategory: Amputations with loss of bone inclusive of one or more joints (not able to be reattached)
LWDC Days: 35
RWDC Days: -
Function: Drilling
Employer: Contractor
Occupation: Transportation operator
Body part: Arm/elbow/shoulder
Nature of injury: Amputation
Time in service: Unspecified

NARRATIVE:

At approximately 12:00 PM, the water truck driver attempted to perform maintenance under the truck with the truck engine running. During the maintenance procedure, the driver's arm was caught between parts of the moving shaft, resulting in a severe injury. The injury was serious enough that the arm was amputated as a consequence of being trapped.

WHAT WENT WRONG:

Investigation is still ongoing.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Investigation is still ongoing.

CAUSAL FACTORS:

No Causal Factors Allocated

DATE: Sep 28 2024

COUNTRY: Nigeria

WORKFORCE PI INJURIES: 1

FUNCTION: Construction

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Construction, commissioning, decommissioning

PRIMARY LIFE-SAVING RULE: Line of fire

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent loss of body parts

PI Subcategory: Amputations with loss of bone inclusive of one or more joints (not able to be reattached)

LWDC Days: -

RWDC Days: -

Function: Construction

Employer: Contractor

Occupation: Process/equipment operator

Body part: Finger(s)/thumb

Nature of injury: Amputation

Time in service: > 6 months < 1 year

NARRATIVE:

At about 00:40, a pipe fitter working on flexible pipelay suffered a hand injury while the crew were attempting to pull the pulling head of the flexible pipe up the J-tube. While trying to stop the pulling head from knocking against adjacent 2" platform diesel pipework, IP's hand got trapped between pulling head and pipework leading to crush amputation of index finger above knuckle. IP was wearing hand gloves at the time of incident.

WHAT WENT WRONG:

N/A

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

N/A

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: Apr 5 2024

COUNTRY: Uganda

WORKFORCE PI INJURIES: 1

FUNCTION: Drilling

CAUSE: Cut, puncture, scrape

ACTIVITY: Office, warehouse, accommodation, catering

PRIMARY LIFE-SAVING RULE: Bypassing safety controls

SECONDARY LIFE-SAVING RULE: Line of fire

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent loss of body parts

PI Subcategory: Amputations with loss of bone inclusive of one or more joints (not able to be reattached)

LWDC Days: 60

RWDC Days: -

Function: Unspecified

Employer: Contractor

Occupation: Other

Body part: Finger(s)/thumb

Nature of injury: Amputation

Time in service: -

NARRATIVE:

Kitchen staff (IP) was using a bone saw machine when the tool skidded off the tabletop. To prevent it from falling, the hands of the IP landed on the blade cutting part of his fingers.

WHAT WENT WRONG:

- Lack of pre-use machine checks. Unsuitable PPE used by the operator during the operation of the machine.
- Relocation of the machine to an unsuitable work bench with smaller dimensions and poor positioning without authorization.
- Unsuitable work surface due to plastic bag-polythene placed between work bench and machine contributing to machine slip.
- Removal of some machine accessories (thickness board and guard) exposing the blades of the saw to operator during use.
- Unacceptable safety act by operator in trying to stabilize the falling/slipping machine using hands to stop it from falling.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

The machine should be repositioned to suitable work bench with ample working surface area. Inspection of tools and equipment should be done prior to use and an evaluation of suitability for task should be done. Comprehensive risk assessment and implementation of daily pre-use operation checklists should be done for all powered kitchen equipment.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

AFRICA OFFSHORE

DATE: Apr 12 2024

COUNTRY: Angola

WORKFORCE PI INJURIES: 1

FUNCTION: Production

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Transport - Water, incl. marine activity

PRIMARY LIFE-SAVING RULE: Line of fire

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent reduction in skin and/or musculoskeletal function

PI Subcategory: Fracture with ongoing impairment to mobility

LWDC Days: 180

RWDC Days: -

Function: Unspecified

Employer: Contractor

Occupation: Other

Body part: Feet/ankle

Nature of injury: Fracture

Time in service: > 6 months < 1 year

NARRATIVE:

During the crew change disembarking from an FPSO to a long range crew boat surfer, the IP was struck by the surfer bow rubber nib while he was positioned between the boat landing ladder and the surfer on ascending wave stairs. As a consequence, the IP suffered a fracture of his right leg ankle.

WHAT WENT WRONG:

- IP still going down on the ladder.
- Seaman instruction was unclear for IP.
- Loudspeaker inefficient.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Devices (loudspeakers) should be checked to assure the use of efficient devices.
- Roles and responsibilities need to be improved.
- Design ladder with bow fender should be compliant with the procedure.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

ASIA/AUSTRALASIA ONSHORE

DATE: May 8 2024
COUNTRY: Australia
WORKFORCE PI INJURIES: 1
FUNCTION: Production
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Line of fire

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent reduction in skin and/or musculoskeletal function
PI Subcategory: Other
LWDC Days: 5
RWDC Days: 262
Function: Production
Employer: Contractor
Occupation: Process/equipment operator
Body part: Finger(s)/thumb
Nature of injury: Cut, Puncture, Scrape
Time in service: -

NARRATIVE:

Injured person received cut to left hand while cutting plywood using circular saw in workshop.

WHAT WENT WRONG:

Tool design does not support the required human action.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

It was identified that the site did not have adequate representation of left-handed battery operated circular saw for the population of users.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

DATE: Nov 4 2024
COUNTRY: Australia
WORKFORCE PI INJURIES: 1
FUNCTION: Production
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Transport - Water, incl. marine activity
PRIMARY LIFE-SAVING RULE: Line of fire
SECONDARY LIFE-SAVING RULE: Unspecified

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent loss of body parts
PI Subcategory: Amputations with loss of bone inclusive of one or more joints (not able to be reattached)
LWDC Days: 114
RWDC Days: 0
Function: Production
Employer: Contractor
Occupation: Other
Body part: Finger(s)/thumb
Nature of injury: Amputation
Time in service: -

NARRATIVE:

During berthing of the vessel, a deckhand experienced a crush injury to their left hand that required surgical amputation of the distal phalanges on the middle and ring fingers.

WHAT WENT WRONG:

Procedural compliance. Distraction. Task planning.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Be wary of operational drift and deviance from following procedures during critical activities. Develop an in-field verification program to monitor line handling practices and associated risks.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Fatigue
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture
PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

ASIA/AUSTRALASIA OFFSHORE

DATE: Feb 17 2024

COUNTRY: Malaysia

WORKFORCE PI INJURIES: 1

FUNCTION: Production

CAUSE: Struck by (not dropped object)

ACTIVITY: Transport - Water, incl. marine activity

PRIMARY LIFE-SAVING RULE: Line of fire

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent loss of body parts

PI Subcategory: Loss of facial structures, example: nose, ears, leading to disfigurement

LWDC Days: -

RWDC Days: -

Function: Production

Employer: Contractor

Occupation: Other

Body part: Head (incl. mouth, etc.)

Nature of injury: Fracture

Time in service: Unspecified

NARRATIVE:

During a hawser connection between the FSO and an offtake tanker, the messenger rope from the hawser became stuck on an assisting vessel's fender. The crew attempted to manoeuvre the assisting vessel away to release the rope, but it snapped back with tension and struck the back of the IP's head. The IP fell forward and hit the vessel's structure (bulwark) which resulted in a facial laceration, jaw fracture, and dental trauma to several teeth.

WHAT WENT WRONG:

Due to current and swell directions, the messenger rope got entangled with the vessel's tyre fenders. As the vessel was clearing the area, the crew was instructed to clear the deck. A crew member (IP), looking for shelter, went under the crash rail, misjudging it was a safe place. The IP noticed tension in the hawser assembly and looked over the bulwark to investigate and at that moment the messenger rope lashed down on the back of the IP's helmet which caused him to fall forward and hit his face against the vessel's bulwark.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Ensure job safety analysis (JSA) for this activity is revised to incorporate safe haven risk of entanglement.
- Reinforce toolbox talk requirements; new shift joiner must perform a toolbox talk prior to entry.
- Refresh all supervisors regarding their supervisory role awareness. Introduce and perform awareness on the requirement of a spotter.
- All parties involved must attend pre-offtake meeting and agree on execution plan (e.g. offtake standard work instructions).
- Tanker escape route, its identification, and activation protocol to be included in the offtake meeting's terms of reference.
- Revisit the FSO offtake Risk Assessment and JSA.
- Review the tyre fender design for operations in the area and explore operational technology advancements for safer messenger rope connections to minimize entanglement risks.

- Incorporate safer hawser connection procedures and 'no slack' requirements into the Standard Work Instructions (SWI).
- FSO ePTW to cover overall offloading activity including review of support vessel and export tanker job hazard analysis (JHA).

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

EUROPE ONSHORE

DATE: Mar 5 2024

COUNTRY: Germany

WORKFORCE PI INJURIES: 1

FUNCTION: Production

CAUSE: Falls from height

ACTIVITY: Maintenance, inspection, testing

PRIMARY LIFE-SAVING RULE: Working at height

SECONARY LIFE-SAVING RULE: Line of fire

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent loss of body parts

PI Subcategory: Loss of internal organs e.g., kidney, spleen (may also meet criteria 3.4.2)

LWDC Days: 365

RWDC Days: -

Function: Production

Employer: Company

Occupation: Maintenance, craftsman

Body part: Internal organs

Nature of injury: Major/multiple system trauma

Time in service: > 5 years < 10 years

NARRATIVE:

During installation work on a sucker rod pump with a cherry picker, the working platform with an employee inside fell approx. 20 ft to the ground.

WHAT WENT WRONG:

Systems - Work Direction or Work Management Issue

- Cleaning/troubleshooting took place concurrently.
 - Work adjacent to equipment without energy isolation.
 - Auto/manual/off/on switch was in the manual position.
 - Troubleshooting led to motor start.

Systems - Work Procedure / Defined Acceptable Practice Issue

- Work adjacent to equipment without energy isolation.
 - Frequency converter switch was in the wrong position for tipping.

People - Lacked Information / Interpretation / Understanding / Communication

- Frequency converter switch was in the wrong position for tipping.

Systems - Work Place Layout / Workplace Design Issue

- Frequency converter switch was in the wrong position for tipping.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

n/a

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

PROCESS (CONDITIONS): Organizational: Inadequate supervision

PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

EUROPE OFFSHORE

DATE: Feb 28 2024

COUNTRY: Norway

WORKFORCE DEATHS: 1

WORKFORCE PI INJURIES: 1

FUNCTION: Production

CAUSE: Aviation accident

ACTIVITY: Transport - Air

PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

WORKFORCE FATALITY

Function: Production

Employer: Company

Occupation: Other

Body part: Respiratory system

Nature of injury: Major/multiple system trauma

Time in service: -

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent reduction in skin and/or musculoskeletal function

PI Subcategory: Unspecified

LWDC Days: -

RWDC Days: -

Function: Production

Employer: Contractor

Occupation: Other

Body part: Neck/torso/spine

Nature of injury: Major/multiple system trauma

Time in service: -

NARRATIVE:

On February 28th, a search and rescue helicopter crashed. The helicopter type was Sikorsky S-92 Search and Rescue (SAR), and it was on a training flight. The accident occurred in connection with training carried out at low altitude in connection with hoisting from the sea. The accident led to the death of the SAR nurse, one of a crew of a total of six people.

WHAT WENT WRONG:

The Norwegian Safety Investigation Authority (NSIA) is still investigating the accident, and no final investigation report is yet available.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

An internal investigation is conducted to improve SAR helicopter safety on the Norwegian Continental Shelf with special focus on emergency response. This investigation recommends measures related to:

- Clearer procedures for when the SAR crew must unfasten the seat belt.
- Review of the personal protective equipment for SAR crew.
- Reintroduce more frequent water training (including helicopter rollovers).
- Use of the same protective equipment during water training as that used in the helicopter.
- More realistic water training within training and drills, protective equipment and coordinated action within the organisation's SAR service and emergency preparedness.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

DATE: Jul 28 2024

COUNTRY: UK

WORKFORCE PI INJURIES: 1

FUNCTION: Production

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Maintenance, inspection, testing

PRIMARY LIFE-SAVING RULE: Energy isolation

SECONARY LIFE-SAVING RULE: Line of fire

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent loss of body parts

PI Subcategory: Amputations with loss of bone inclusive of one or more joints (not able to be reattached)

LWDC Days: 13

RWDC Days: 58

Function: Production

Employer: Contractor

Occupation: Maintenance, craftsman

Body part: Finger(s)/thumb

Nature of injury: Amputation

Time in service: Unspecified

NARRATIVE:

During non-invasive temperature monitoring of the west crane engine, as part of repair commissioning, the engineer observed an oil filter leaking (this is a known leak). Whilst investigating the leak the IP placed his hand in an area not protected by the radiator fan guard (engine running) resulting in the hand making contact with the radiator fan resulting in significant injury to fingers of his left-hand and required Medevac to hospital.

WHAT WENT WRONG:

Lack of risk awareness.

Lack of spatial awareness.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Energy isolation.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

MIDDLE EAST ONSHORE

DATE: May 12 2024

COUNTRY: Kuwait

WORKFORCE PI INJURIES: 1

FUNCTION: Drilling

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Drilling, workover, well operations

PRIMARY LIFE-SAVING RULE: Bypassing safety controls

SECONDARY LIFE-SAVING RULE: Line of fire

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent loss of body parts

PI Subcategory: Amputations with loss of bone inclusive of one or more joints (not able to be reattached)

LWDC Days: -

RWDC Days: -

Function: Drilling

Employer: Contractor

Occupation: Drilling/well servicing operator

Body part: Leg/knee/hip

Nature of injury: Amputation

Time in service: -

NARRATIVE:

While drilling was in progress, the Assistant Driller (A/D) noticed a blockage of drill cuttings near feeding port cover plate in the extended portion of the auger. The A/D came near the cover plate to flush the accumulated drill cuttings with water. As the water hose was on other side of the auger, the AD tried to climb over the cover plate to reach that side. When he was trying to crossover putting his right foot on the cover plate, his right leg slipped into the running auger as the cover plate was not secured with bolts. A Roustabout switched off the auger on noticing the incident. However, by that time, the AD's right foot was already severed below the ankle with the amputated portion stuck in the auger blade.

WHAT WENT WRONG:

- The metal plate (located in the middle of one of the gratings) had not been linked with auger interlock system and was not properly secured.
- Specified controls by the Company not followed by the Contractor.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- All hazards and safety barriers must be sustained with adherence to procedures.
- All safety barriers must be sustained, including to secure the Cover Plate through bolting, safety interlocks and extension of water jet system.

CAUSAL FACTORS:

No Causal Factors Allocated

DATE: Nov 23 2024

COUNTRY: Kuwait

WORKFORCE PI INJURIES: 1

FUNCTION: Unspecified

CAUSE: Cut, puncture, scrape

ACTIVITY: Drilling, workover, well operations

PRIMARY LIFE-SAVING RULE: Line of fire

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent loss of body parts

PI Subcategory: Amputations with loss of bone inclusive of one or more joints (not able to be reattached)

LWDC Days: 90

RWDC Days: 0

Function: Drilling

Employer: Contractor

Occupation: Drilling/well servicing operator

Body part: Finger(s)/thumb

Nature of injury: Amputation

Time in service: > 5 years < 10 years

NARRATIVE:

During the retrieval of the Electrical Submersible Pump (ESP) assembly including its power cable, the bottom of the pump section of the ESP parted below the rig floor, just above the flange. This occurred while two workers were removing cable guards, bands, and disposing of the 1/4" capillary tubing. The lower pump portion fell back into the well, and the power cable parted, striking one worker. The second worker was manually handling the capillary tubing attached to the bottom of the ESP assembly, which wrapped around his hand and severed his left thumb.

WHAT WENT WRONG:

Potential consequences not understood: In an effort to "help," the rig floor had more people on the rig floor, placing people in the line of fire. Hazard not recognized: The risk of the pump failure pulling the IP was not recognized. The risk of an ESP failure from corrosion was not identified. Inadequate resource allocation: An issue with the inability to connect a chemical injection skid to a high-voltage ESP well was identified, but a solution was not implemented. The ESP is disassembled section by section for ~6 hours; this leaves the rig with a key well control safeguard during the operation.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

N/A

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: Dec 14 2024
COUNTRY: Oman
WORKFORCE PI INJURIES: 1
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Line of fire
SECONDARY LIFE-SAVING RULE: Unspecified

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent reduction of organ function
PI Subcategory: Eye injury with loss of vision
LWDC Days: 73
RWDC Days: 0
Function: Production
Employer: Contractor
Occupation: Manual labourer
Body part: Eye(s)
Nature of injury: Sensory loss/damage (e.g., hearing, sight, taste, smell)
Time in service: -

NARRATIVE:

While the pipe spools were being loaded into the HIAB truck using the Hydraulic Telescopic forklift forks, one of the pipe spools was pushed by the Rigger and HIAB operator from the forklift. The pipe dropped onto the HIAB truck and struck a piece of wood, which bounced and hit the Rigger's face.

WHAT WENT WRONG:

Investigation not finalized.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Always ensure the team are following Life-Saving Rules with particular to line of fire and work authorizations.

CAUSAL FACTORS:

No Causal Factors Allocated

DATE: Apr 14 2024
COUNTRY: Oman
WORKFORCE PI INJURIES: 1
FUNCTION: Unspecified
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Line of fire

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent loss of body parts
PI Subcategory: Loss of eye(s)
LWDC Days: 77
RWDC Days: 0
Function: Unspecified
Employer: Contractor
Occupation: Drilling/well servicing operator
Body part: Eye(s)
Nature of injury: Cut, Puncture, Scrape
Time in service: > 1 year < 5 years

NARRATIVE:

At approximately 09:00 hours on Sunday, April 14th, while pulling out of hole (POOH) with a stuck insert pump, a 2 foot pony rod dislodged from its coupling, fell 20 feet, and struck a contractor, resulting in multiple injuries to his face and eye.

Immediate medical attention was provided, with the IP eventually transferred to hospital for surgery.

WHAT WENT WRONG:

Dropped object.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Failure to follow procedure.

CAUSAL FACTORS:

No Causal Factors Allocated

DATE: Jan 21 2024
COUNTRY: Qatar
WORKFORCE PI INJURIES: 1
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Line of fire
SECONARY LIFE-SAVING RULE: Working at height

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent loss of body parts
PI Subcategory: Loss of internal organs e.g., kidney, spleen
LWDC Days: 143
RWDC Days: 0
Function: Production
Employer: Contractor
Occupation: Manual labourer
Body part: Internal organs
Nature of injury: Organ function / loss or damage
Time in service: > 5 years < 10 years

NARRATIVE:

During one of the scaffolding material demobilizations from higher elevation by using handballing methodology, a scaffolding cup lock ledger (9kg) slipped from one of the Scaffolders at elevation.

The ledger fell outside the scaffold structure and hit a nearby staircase hand railing. This changed the fall direction and gradually hit another Scaffolders who was at the 4th level of the same scaffold structure.

He was sent to the clinic for First Aid, and then to Hospital.

Scaffold activities by relevant party are suspended and investigation is kicked off.

WHAT WENT WRONG:

Scaffolders' moved Scaffolders Collision of ledgers due to unexpected change of pace of moving ledgers.

Deflection of falling ledger by nearby staircase into Scaffolders's section.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Communication among work parties during work execution and moments of change.
- Clarity of pace of movement when handballing.
- Planning of work to mitigate collision and drops during material movement at height.
- Selection of access method for working at height.
- Managing drop risks during scaffold ledger movement.
- Location of scaffold structure in proximity to structures; protection of workers inside scaffold section.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Protective Systems: Inadequate security provisions or systems

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

DATE: Feb 20 2024

COUNTRY: Qatar

WORKFORCE PI INJURIES: 1

FUNCTION: Construction

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Construction, commissioning, decommissioning

PRIMARY LIFE-SAVING RULE: Line of fire

SECONDARY LIFE-SAVING RULE: Energy isolation

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent loss of body parts

PI Subcategory: Amputations with loss of bone inclusive of one or more joints (not able to be reattached)

LWDC Days: 25

RWDC Days: 6

Function: Construction

Employer: Contractor

Occupation: Foreman, Supervisor

Body part: Finger(s)/thumb

Nature of injury: Crushing

Time in service: > 1 year < 5 years

NARRATIVE:

Finger crushed in truck belt pulley system. During repair and maintenance, the Injured Person (IP) reinstalled a bracket support for a truck alternator and air conditioner (AC) compressor. After completing the reinstallation, the engine was started to check the alignment of the belt pulley system attached to the alternator and AC compressor. The IP removed his glove from his left hand to check the temperature of the AC compressor by feeling the inlet and discharge pipes with the back of his hand while the engine was running. While doing so, the IP's hand got caught in the belt and was dragged through the pulley, crushing his finger.

WHAT WENT WRONG:

- The IP placed his hand in the Line of Fire.
- The equipment was running while maintenance work was being performed.
- The wrong tool was used for the job.

What Worked?:

The IP was taken to a Medical Aid Centre and then to Hospital for further assessment and medical management.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Always be aware of your hand and body placement while working with rotating equipment, and never place your hand in the Line of Fire.
- Do not attempt to conduct maintenance or repairs on equipment while it is running.
- Use the correct tool for the job.
- Be mindful of pinch points and possible hazards or risks that may place you in the Line of Fire.

CAUSAL FACTORS:

No Causal Factors Allocated

DATE: Sep 25 2024

COUNTRY: Qatar

WORKFORCE PI INJURIES: 1

FUNCTION: Construction

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Construction, commissioning, decommissioning

PRIMARY LIFE-SAVING RULE: Energy isolation

SECONDARY LIFE-SAVING RULE: Line of fire

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent loss of body parts

PI Subcategory: Amputations with loss of bone inclusive of one or more joints (not able to be reattached)

LWDC Days: 82

RWDC Days: -

Function: Construction

Employer: Contractor

Occupation: Foreman, Supervisor

Body part: Finger(s)/thumb

Nature of injury: Amputation

Time in service: > 5 years < 10 years

NARRATIVE:

Multiple Finger Injuries: Wind Moves Impeller of Blower Belt and Pulley System.

An Electrical Foreman (Injured Person 1 / IP1) was removing a frayed thread from a blower fan belt when a gust of wind moved the impeller, trapping his finger between the pulley and belt. The SHES Manager (IP2) went to evaluate the situation, checked the belt tension in the same place, and also sustained finger injuries. The blower was not connected to any power source at the time.

WHAT WENT WRONG:

The belt and pulley had a guard, but it was insufficient, leading both individuals to place their hands on the belt. They assumed no potential energy was present since the blower was not powered.

- A sudden gust of wind rotated the air blower impeller, which moved the belt and pulley.
- The risk of wind affecting the blower was not anticipated in project procedures, including the Energy Isolation Program (LOTO), Method Statement, and Job Safety Analysis. The equipment manufacturer only advised isolating the power source, which both individuals assumed was sufficient.

What worked?:

Case management was followed.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Plan for all energy sources, including electric, hydraulic, pneumatic, spring-loaded, gravity, and wind.
- Ensure ZERO energy before starting any work, such as inspection or maintenance.
- Never place body parts into rotating equipment (Line of Fire); ensure guards and signs are in place.
- Conduct hazard identification and mitigation before entering the incident scene for data collection.
- Discuss solutions with the blower manufacturer to prevent impeller rotation during installation and maintenance.
- Update the energy isolation program and training to include incident findings and emphasise “Lock, Block & Bleed.”
- Communicate incident findings to all site personnel during Toolbox Talks.

CAUSAL FACTORS:

No Causal Factors Allocated

DATE: Jan 11 2024
COUNTRY: UAE
WORKFORCE PI INJURIES: 1
FUNCTION: Drilling
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Line of fire

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent loss of body parts
PI Subcategory: Amputations with loss of bone inclusive of one or more joints (not able to be reattached)
LWDC Days: -
RWDC Days: -
Function: Drilling
Employer: Contractor
Occupation: Manual labourer
Body part: Finger(s)/thumb
Nature of injury: Crushing
Time in service: > 1 year < 5 years

NARRATIVE:

Finger injury: During P00H, while pulling hawk jaw towards drill pipe (DP) stand to break-out, the Assistant Driller sustained injuries in his right hand middle and index fingers due to caught in-between DP stand and the hawk jaw control panel cover. The injured person was provided first aid by rig medic and sent to site clinic; from there he was referred to hospital for further assessment. Classified on 13th Feb 2024.

WHAT WENT WRONG:

Inadequate hazard identification or risk assessment.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not available.

CAUSAL FACTORS:

No Causal Factors Allocated

MIDDLE EAST OFFSHORE

DATE: Jun 28 2024

COUNTRY: Qatar

WORKFORCE PI INJURIES: 1

FUNCTION: Production

CAUSE: Explosion, fire or burns

ACTIVITY: Production operations

PRIMARY LIFE-SAVING RULE: Line of fire

SECONARY LIFE-SAVING RULE: Work authorization

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent reduction of organ function

PI Subcategory: Genital or other sexual organ injury leading to loss of function

LWDC Days: 63

RWDC Days: -

Function: Production

Employer: Company

Occupation: Foreman, Supervisor

Body part: Genitalia or other sexual organs

Nature of injury: Unspecified

Time in service: > 1 year < 5 years

NARRATIVE:

Electro-chlorinator cell-A cover blown off causing multiple injuries. Following a corrective maintenance activity carried out during the day, the programmable logic controller (PLC) of the electro-chlorination unit was reset. After the reset, a violent explosion occurred in the electro-chlorination unit, causing a blast and multiple injuries to the injured person (IP) who was in the immediate vicinity. The general platform alarm was initiated, and emergency protocols were followed. The IP was found lying on the deck. He was transferred to the clinic, and then safely evacuated to hospital by Medevac chopper.

WHAT WENT WRONG:

- Respect Hazards: Hydrogen detection was not adequately addressed in the current design.
- Apply Procedures: Operating and maintenance practices did not follow the recommendations from the original equipment manufacturer (OEM). There were gaps in the application of Company procedures (e.g., Permit to Work).
- Watch for Weak Signals: There was a failure to apply executive action in response to the observed degradation of various systems (Management of Change, quality control, surveillance, emergency management, and the Permit to Work system).
- Maintain Competency: There was an absence of Hydrogen risk awareness among offshore operations personnel.
- Manage Change: No risk assessment or HAZID/HAZOP was carried out within the framework of the eMOC, and a last-minute change was implemented without following the eMOC procedure.

What went well?:

Emergency response was immediately initiated, and case management was followed.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Ensure that all steps of the MOC process are effectively applied and that all expected deliverables are issued at the required level of quality.
- Verify that any hydrogen risk is adequately addressed in the safety reviews of electro-chlorination units,

including the limitation of personnel exposure.

- Verify the adequacy of the equipment strategy with the vendor's recommendations.
- Ensure that executive action is taken when degradation in site conditions, operating parameters, or procedures is observed.

CAUSAL FACTORS:

No Causal Factors Allocated

NORTH AMERICA ONSHORE

DATE: Sep 29 2024

COUNTRY: Canada

WORKFORCE PI INJURIES: 1

FUNCTION: Drilling

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Lifting, crane, rigging, deck operations

PRIMARY LIFE-SAVING RULE: Line of fire

SECONARY LIFE-SAVING RULE: Safe mechanical lifting

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent loss of body parts

PI Subcategory: Amputations with loss of bone inclusive of one or more joints (not able to be reattached)

LWDC Days: 123

RWDC Days: 30

Function: Drilling

Employer: Contractor

Occupation: Transportation operator

Body part: Finger(s)/thumb

Nature of injury: Amputation

Time in service: 10 + years

NARRATIVE:

A bed truck was positioning mats for a future tank. The injured worker (IP), who was the driver of a second bed truck and assisting as a swamper, moved the winch line laterally on the live roll, a cylindrical component that allows the winch line to move and adjust the load's position. As the bed truck operator engaged the winch to align the mats, the IP placed both hands under the lifting sling near the live roll. This caused the IP's fingers to get caught, resulting in injuries to both hands. The IP was transported to a hospital for treatment.

WHAT WENT WRONG:

The injured worker placed both hands under the lifting sling near the live roll while the bed truck operator engaged the winch to align the mats.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

The contracting company lacked a documented procedure for re-positioning matting and did not have a written expectation for maintaining a safe distance from winch lines or loads in motion (exclusion zone).

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: Mar 8 2024

COUNTRY: USA

WORKFORCE PI INJURIES: 1

FUNCTION: Drilling

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Drilling, workover, well operations

PRIMARY LIFE-SAVING RULE: Line of fire

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent loss of body parts

PI Subcategory: Amputations with loss of bone inclusive of one or more joints (not able to be reattached)

LWDC Days: -

RWDC Days: -

Function: Drilling

Employer: Contractor

Occupation: Drilling/well servicing operator

Body part: Finger(s)/thumb

Nature of injury: Amputation

Time in service: Unspecified

NARRATIVE:

IP was strapping directional tools that were on pipe racks. While strapping a mud motor, to get a better view of the scribe line, he rolled the motor until the bend was in the uppermost position, or "High Sided" it. Seeking stability, the motor then rolled bend-down and onto IP's left thumb on the pipe rack. Further medical assessment resulted in a determination to amputate the distal phalanx of the thumb.

WHAT WENT WRONG:

Failure to utilize proper tools (i.e., pipe handling bar) to move mud motor.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Workers now required to ensure use of pipe handling equipment when moving pipe for strapping.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PEOPLE (ACTS): Use of Protective Methods: Personal Protective Equipment not used or used improperly

PEOPLE (ACTS): Use of Protective Methods: Disabled or removed guards, warning systems or safety devices

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: Nov 21 2024

COUNTRY: USA

WORKFORCE PI INJURIES: 1

FUNCTION: Drilling

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Drilling, workover, well operations

PRIMARY LIFE-SAVING RULE: Line of fire

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent loss of body parts

PI Subcategory: Amputations with loss of bone inclusive of one or more joints (not able to be reattached)

LWDC Days: -

RWDC Days: -

Function: Drilling

Employer: Contractor

Occupation: Drilling/well servicing operator

Body part: Finger(s)/thumb

Nature of injury: Amputation

Time in service: Unspecified

NARRATIVE:

Severe Injury occurred while coil tubing unit crew was rigging down and securing injector onto cradle for transport. Injured person (IP) placed his left hand in rear-guide pin hole of cradle to support himself as he made his way to the back of the unit. Pinch point incident resulted in amputation of IP's middle finger when injector shifted slightly as it settled into position.

WHAT WENT WRONG:

- Improper Hand Placement: IP failed to identify pinch points.
- Inadequate Job Safety Analysis: Crew had change of scope shortly after shift began and did not regroup or complete new JSA for activity that was taking place.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Contractor to ensure mechanical changes implemented to reduce risk of exposed pin-hole. Re-training on JSA completion and updates when job scope changes.

CAUSAL FACTORS:

PROCESS [CONDITIONS]: Organizational: Inadequate hazard identification or risk assessment

DATE: Feb 4 2024
COUNTRY: USA
WORKFORCE PI INJURIES: 2
FUNCTION: Production
CAUSE: Explosion, fire or burns
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Line of fire

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent reduction in skin and/or musculoskeletal function
PI Subcategory: Burns: deep burns with scarring or skin grafts that impair body movement or other function
LWDC Days: -
RWDC Days: -
Function: Production
Employer: Contractor
Occupation: Drilling/well servicing operator
Body part: Head (incl. mouth, etc.)
Nature of injury: Burn
Time in service: -

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent reduction in skin and/or musculoskeletal function
PI Subcategory: Burns: deep burns with scarring or skin grafts that impair body movement or other function
LWDC Days: -
RWDC Days: -
Function: Production
Employer: Contractor
Occupation: Drilling/well servicing operator
Body part: Head (incl. mouth, etc.)
Nature of injury: Burn
Time in service: -

NARRATIVE:

While drilling out frac plugs, the flowback manifold was aligned to flow through the mud-gas separator to flare. Gas was flared for approximately 5 minutes before a flash fire occurred near the mud-gas separator. Three crew members sustained burns as a result of the fire.

WHAT WENT WRONG:

Contact by - Temperature Extremes

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

N/A

CAUSAL FACTORS:

No Causal Factors Allocated

DATE: Oct 9 2024

COUNTRY: USA

WORKFORCE PI INJURIES: 1

FUNCTION: Production

CAUSE: Explosion, fire or burns

ACTIVITY: Production operations

PRIMARY LIFE-SAVING RULE: Bypassing safety controls

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent reduction in skin and/or musculoskeletal function

PI Subcategory: Burns: deep burns with scarring or skin grafts that impair body movement or other function

LWDC Days: -

RWDC Days: -

Function: Production

Employer: Contractor

Occupation: Manual labourer

Body part: Neck/torso/spine

Nature of injury: Burn

Time in service: Unspecified

NARRATIVE:

Contractors arrived at the facility to empty and clean 6 storage tanks as part of the area tankless conversion project. The contracted equipment package includes a 3 man team with a hydrovac truck and 1 man with a vacuum truck. The crew completed their JSA and forwarded it to the site supervisor for approval. The IC's started work on the east 500 barrel produced water tank. They valved out the fluid in the tank with the vacuum truck and proceeded to pull the manway hatch on the back of the tank. The hydrovac truck was used to finish cleaning the tank with a 4 inch suction hose. They finished cleaning the produced water tank and successfully cleaned a second produced water tank in the same fashion. The crew moved the equipment down to the next tank, a 500 barrel oil storage tank, and valved the fluid out as they did the produced water tanks. They removed the manway hatch and used the 4 inch suction hose to vacuum out the basic sediment and water that remained behind. After a short time, the swamper and vacuum truck driver positioned themselves to offload the fluid in the hydrovac to the tanker of the vacuum truck. The hydrovac operator/supervisor continued to manoeuvre the hose in the oil tank while the swamper was going to pump into the vacuum truck. At this time, the oil storage tank exploded and sent a flash fire through the manway opening which burned all 4 contractors.

WHAT WENT WRONG:

The crew did not assure proper equipment grounding prior to the job. A static discharge is believed to be the primary cause of ignition.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Overall process for cleaning out tanks was revised. Proper grounding must be assured, and additional supervision is now required.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing
PROCESS (CONDITIONS): Organizational: Inadequate training/competence
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: Oct 28 2024

COUNTRY: USA

WORKFORCE PI INJURIES: 1

FUNCTION: Construction

CAUSE: Exposure electrical

ACTIVITY: Maintenance, inspection, testing

PRIMARY LIFE-SAVING RULE: Energy isolation

SECONARY LIFE-SAVING RULE: Line of fire

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent reduction in skin and/or musculoskeletal function

PI Subcategory: Traumatic soft tissue injury to shoulder/knee/elbow with ongoing pain and/or inadequate joint range of movement

LWDC Days: 63

RWDC Days: 0

Function: Construction

Employer: Contractor

Occupation: Maintenance, craftsman

Body part: Arm/elbow/shoulder

Nature of injury: Burn

Time in service: > 5 years < 10 years

NARRATIVE:

IP was conducting work on power distribution lines utilizing bucket truck to reach elevated power line equipment. The IP's left shoulder contacted a live power line energized with approximately 7200 volts. The electricity travelled down IP's left arm and exiting IP's left wrist.

WHAT WENT WRONG:

Legal hold

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

n/a

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

DATE: Feb 16 2024

COUNTRY: USA

WORKFORCE PI INJURIES: 1

FUNCTION: Unspecified

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Drilling, workover, well operations

PRIMARY LIFE-SAVING RULE: Line of fire

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent loss of body parts

PI Subcategory: Amputations with loss of bone inclusive of one or more joints (not able to be reattached)

LWDC Days: 7

RWDC Days: 55

Function: Unspecified

Employer: Contractor

Occupation: Drilling/well servicing operator

Body part: Finger(s)/thumb

Nature of injury: Amputation

Time in service: > 1 year < 5 years

NARRATIVE:

Contractor at approximately 10:45 AM Friday February 16, at a well servicing rig was running rods in the hole. The rig blocks were traveling up and struck the rod basket causing the hinged front section to lift. As the hinged section returned, the IP's left ring finger was caught in the rod basket hinge. The IP was transported for medical treatment. The injury resulted in the amputation of the distal joint of the IP's left ring finger.

WHAT WENT WRONG:

Line of fire.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Failure to follow procedure.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate communication

DATE: Nov 24 2024

COUNTRY: USA

WORKFORCE PI INJURIES: 1

FUNCTION: Unspecified

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Lifting, crane, rigging, deck operations

PRIMARY LIFE-SAVING RULE: Line of fire

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent loss of body parts

PI Subcategory: Amputations with loss of bone inclusive of one or more joints (not able to be reattached)

LWDC Days: 10

RWDC Days: -

Function: Unspecified

Employer: Contractor

Occupation: Drilling/well servicing operator

Body part: Finger(s)/thumb

Nature of injury: Amputation

Time in service: > 3 months < 6 months

NARRATIVE:

A drilling contractor employee (injured person – IP) was attempting to lift a 6ft sub weighing approximately 150 pounds into a scrap bin when he lost control of the sub and pinched the middle finger on his left hand between the sub and a butterfly valve that was located in the scrap bin. The IP was transported to hospital where he received treatment for a partial finger amputation.

WHAT WENT WRONG:

Line of fire.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Evaluate hazard and develop a plan of action.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate supervision

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

NORTH AMERICA OFFSHORE

DATE: Mar 30 2024

COUNTRY: USA

WORKFORCE PI INJURIES: 1

FUNCTION: Drilling

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Maintenance, inspection, testing

PRIMARY LIFE-SAVING RULE: Line of fire

SECONDARY LIFE-SAVING RULE: Unspecified

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent loss of body parts

PI Subcategory: Amputations with loss of bone inclusive of one or more joints (not able to be reattached)

LWDC Days: 101

RWDC Days: 0

Function: Drilling

Employer: Contractor

Occupation: Foreman, Supervisor

Body part: Finger(s)/thumb

Nature of injury: Amputation

Time in service: -

NARRATIVE:

Workers were installing handrails in an upper level of the port riser bay. A worker on an upper level was positioned where the handrail would be lifted and welded in place. While the handrail section was being hoisted the individual lost balance and instinctively grabbed a pneumatic hoist wire rope with the right hand to regain balance, the right thumb was pulled into the sheave resulting in significant injury.

The worker was evaluated by the rig medic and after consult by the onshore physician, the worker was sent ashore by Medevac. The worker departed the rig.

WHAT WENT WRONG:

The causal investigation identified two potential theories related to how the worker's thumb was pulled into the sheave. Theory 1: When the event happened, it's possible that the worker lost footing and grabbed the wire rope to steady themselves. Theory 2: The worker was guiding the wire rope when their gloved thumb contacted the sheave.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Key Lessons learned were related to fatigue risk management for third-party contractors, sharing is not the same as learning (a similar event occurred in 2021 that did not result in a FPI event). Elements of human performance played a significant role in this event - fatigue, perceived pressure to complete the work, and people (including the IP) were performing multiple roles at the same time which created distraction.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

RUSSIA & CENTRAL ASIA ONSHORE

DATE: Jan 10 2024

COUNTRY: Azerbaijan

WORKFORCE PI INJURIES: 1

FUNCTION: Unspecified

CAUSE: Struck by (not dropped object)

ACTIVITY: Transport - Land

PRIMARY LIFE-SAVING RULE: Driving

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent loss of body parts

PI Subcategory: Amputations with loss of bone inclusive of one or more joints (not able to be reattached)

LWDC Days: 180

RWDC Days: 0

Function: Unspecified

Employer: Company

Occupation: Transportation operator

Body part: Leg/knee/hip

Nature of injury: Amputation

Time in service: > 5 years < 10 years

NARRATIVE:

A shuttle bus driver was returning to the garage after dropping off workers. On the road the bus collided with a trailer parked along the side of the road. The impact severely damaged the front of the bus and resulted in serious injuries to the driver's lower body. The driver was quickly transported to a trauma hospital by an ambulance team, where both of his legs were amputated below the knee due to the extent of his injuries.

WHAT WENT WRONG:

- Lack of fatigue management procedure.
- Inadequate medical screening of the drivers.
- Inadequate training programme.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Situational awareness skills of the drivers to be improved.
- Defensive driving training courses to be provided.
- Fatigue management procedure to be implemented.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

RUSSIA & CENTRAL ASIA OFFSHORE

No PI incidents reported

SOUTH & CENTRAL AMERICA ONSHORE

DATE: Aug 14 2024

COUNTRY: Argentina

WORKFORCE PI INJURIES: 1

FUNCTION: Drilling

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Drilling, workover, well operations

PRIMARY LIFE-SAVING RULE: Line of fire

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent loss of body parts

PI Subcategory: Amputations with loss of bone inclusive of one or more joints (not able to be reattached)

LWDC Days: 139

RWDC Days: 0

Function: Drilling

Employer: Contractor

Occupation: Manual labourer

Body part: Toes

Nature of injury: Amputation

Time in service: -

NARRATIVE:

During rig up rig down tasks, while disassembling the hydraulic seal of the striker, an 8" joint was disconnected, causing the hydraulic seal (U-shaped pipe) to slide down and trap the operator's left foot.

WHAT WENT WRONG:

U-shaped pipe slid down and trapped the operator's left foot.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Maintain a high sense of vulnerability.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: Oct 5 2024

COUNTRY: Argentina

WORKFORCE PI INJURIES: 1

FUNCTION: Construction

CAUSE: Struck by (not dropped object)

ACTIVITY: Construction, commissioning, decommissioning

PRIMARY LIFE-SAVING RULE: Energy isolation

SECONDARY LIFE-SAVING RULE: Line of fire

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent reduction in psychological, social or cognitive function

PI Subcategory: Brain injury with ongoing impact to emotional regulation, speech, memory, or other cognitive dysfunction

LWDC Days: 147

RWDC Days: -

Function: Construction

Employer: Contractor

Occupation: Foreman, Supervisor

Body part: Head (incl. mouth, etc.)

Nature of injury: Concussion

Time in service: > 1 year < 5 years

NARRATIVE:

Struck on the head by a 4 inch hose due to a sudden pressure release.

WHAT WENT WRONG:

- No management of change implemented to validate the facility modifications.
- The contractor company had no experience in air blowing the separator pipe.
- Operator on the line of fire.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Update the Piping Class including hoses for different uses.
- Review supervision personnel competencies (LTO).
- Reinforce the Management of change process.

CAUSAL FACTORS:

No Causal Factors Allocated

SOUTH & CENTRAL AMERICA OFFSHORE

DATE: Feb 9 2024

COUNTRY: Brazil

WORKFORCE PI INJURIES: 1

FUNCTION: Drilling

CAUSE: Confined space

ACTIVITY: Maintenance, inspection, testing

PRIMARY LIFE-SAVING RULE: Line of fire

SECONDARY LIFE-SAVING RULE: Confined space

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent reduction in psychological, social or cognitive function

PI Subcategory: Other

LWDC Days: 27

RWDC Days: -

Function: Drilling

Employer: Contractor

Occupation: Heavy equipment operator

Body part: Neck/torso/spine

Nature of injury: Crushing

Time in service: > 5 years < 10 years

NARRATIVE:

During an operational manoeuvre to deballast a semisubmersible rig and reduce its draft—allowing it to be docked at the shipyard for the removal of coral sol—it was planned to drain the dead volume from the ballast tanks. After deballasting, approximately 5% of the tank's total capacity remained as "dead volume." To remove this, a stripping operation was performed, which involves opening the ballast tank's manhole cover and, if necessary, carrying out confined space work to insert the suction hose of a portable pneumatic pump to extract the remaining water.

To access Tank #6 in Column "E" the manhole cover was removed, revealing a continuous rubber gasket used as a sealing element, which was still completely blocking the manhole entry. Upon removing the rubber gasket, a pressure differential was present (Tank #6 had lower internal pressure than the pump room), causing two of the three workers involved to be sucked in, colliding with the manhole structure and becoming trapped due to the pressure difference. The incident resulted in severe injuries—one worker sustained an injury to the right side of the body, while the other suffered a cut on the upper lip, bleeding, and loss of consciousness.

WHAT WENT WRONG:

The increase in the rig's load in 2006, amounting to several tons, created the need to open the manhole for stripping operations (removal of dead volume) to achieve the unit's minimum draft, a necessary condition for docking manoeuvres.

- The construction characteristic of the sealing element (a continuous rubber gasket) covered the entire cross-section of the manhole, violating ASME B16.21 standards, thereby isolating the internal and external environments even after the cover was removed. This arrangement required the worker to break the rubber gasket to access the tank.
- The vent of Tank #6 was undersized for deballasting manoeuvres using two pumps (main and backup) operating simultaneously.
- The manhole of Tank #6 was opened while another tank from the same column was being deballasted. Tanks in the same column share a common suction header with the ballast pumps.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Portable Pump Stripping

- Utilize a portable pump for dead volume removal.

Compliance with ASME B16.21

- Ensure sealing gaskets for ellipses and manholes comply with ASME B16.21 requirements.

Simultaneous Operations:

- Assess whether simultaneous deballasting with two pumps is allowed and revise the Operations Manual if necessary.
- Prohibit access to the ballast tank (manhole opening) while another tank in the same column is being pumped out (deballasting).

Energy Isolation – Evaluate the Lockout/Tagout (LOTO) Procedure to include the following:

- Electrical lockout of process equipment (e.g., pump motors) for interventions in non-electrical work zones (e.g., flange opening).
- Lockout of remotely operated valves (e.g., XV, HV), preventing inadvertent operation.
- Lockout form completion requirements, highlighting the need to identify all control devices/blocking points required to manage all hazardous energy sources related to the equipment intervention.
- Allowing the “double-check” of isolation to be performed by another authorized employee responsible for the system/equipment.
- Pipeline isolation for confined spaces, ensuring all pipelines leading to the confined space are isolated with a blind flange or spectacle blind as close as possible to the confined space.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

DATE: May 24 2024

COUNTRY: Trinidad and Tobago

WORKFORCE PI INJURIES: 1

FUNCTION: Drilling

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Drilling, workover, well operations

PRIMARY LIFE-SAVING RULE: Line of fire

SECONARY LIFE-SAVING RULE: Work authorization

WORKFORCE PERMANENT IMPAIRMENT

PI Category: Permanent loss of body parts

PI Subcategory: Amputations with loss of bone inclusive of one or more joints (not able to be reattached)

LWDC Days: -

RWDC Days: -

Function: Drilling

Employer: Contractor

Occupation: Drilling/well servicing operator

Body part: Finger(s)/thumb

Nature of injury: Amputation

Time in service: -

NARRATIVE:

Five crew members were in the process of measuring 13 5/8" casing joints on the main deck of the drilling rig. Three crew members were on the port side and two on the starboard side. The casing was placed on top of the rack and spread out. The port side crew members used a rope to roll one joint at a time towards the aft of the rig with the assistance of the starboard crew member who pushed the pipe to give it a start. Once a joint was stopped, a crew member on either side would back the thread protectors off before rolling the next joint of casing. The injured person (IP) was on the starboard side backing out the box protector on one of the casing joints when another casing joint was rolled toward him. The IP's right index finger was caught between the two joints of casing. His injury resulted in a medevac and subsequent amputation of the finger.

WHAT WENT WRONG:

The work instruction did not adequately detail steps for the job and the risks of rolling joints and removing casing protectors were not captured; the crew attempted to improve the process of strapping joints using rope, however, the risks introduced by this method were not adequately assessed. The casing had been previously measured onshore. Due to inaccuracies seen in previous onshore measurements a decision was made to strap every joint offshore at the rig-site. A breakdown in communication led to IP's hands being in the Line of Fire. He was in the process of backing off the box protector using his hands when another crew member rolled the casing onto his hand.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Critical review of lower risk activities on sites which can result in life changing injury (LCI)/fatalities
2. Eliminate/minimize normalization of hand injury line of fire risks.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

This page is intentionally blank



IOGP Headquarters

3 Moorgate Place, London, EC2R 6EA, United Kingdom
T: +44 20 4570 6879
E: reception@iogp.org

IOGP Europe

T: +32 2 882 16 53
E: reception-europe@iogp.org

www.iogp.org